An Introduction to Isogeny-based Cryptography

CONTENTS

INTRODUCTION TO THE BOOK

The field of isogeny-based cryptography is rapidly expanding since the 2020s. To keep up with developments in isogeny-based cryptography, we need a strong understanding of the basics, and a gentle introduction to advanced topics. Unfortunately, there is no unified place yet, where anyone can go to learn these topics, nor is there unified notation or sometimes even definitions between authors and works.

This collaborative open-source text book aims to fill that gap by providing introductory materials to all topics in isogeny-based cryptography. We strongly focus on intuition and examples to grasp these materials, to ensure that we turn mathematical insight into concrete cryptographic applications.

We strive to describe every aspect of isogeny-based cryptography, with ample references to all related work that introduced or expanded on these concepts. It will be impossible to keep up with the current progress in isogeny-based cryptography; whenever we have not yet adequately described a topic, we simply list the literature to provide a starting point for the reader.

THE MATERIAL. We hope to cover all of the basics, and most of the advanced material required to understand modern isogeny-based cryptography. However, this is a daunting task. Whenever a chapter on something is planned, but not yet written, we will refer to materials that are either an introduction to or a comprehensive treatment of the subject.

THE READER. The reader is a somewhat mystical entity, who has certain skills but lacks other skills, has certain needs for which they have come to this book, and have a certain attention span. In general, this book tries to give you the information you need while keeping their explanations as elementary as possible.

We will often skip or delay proofs. This is an unfortunate side-effect that results from the fact that elliptic curves and isogenies have beautiful theorems and results that arise from deep(er) mathematics. So, often, we want to discuss a certain elementary statement about curves and isogenies, even though the proof requires material that will only be discussed much later. As an example, take Falting's Theorem:

Theorem 1. Let C be a smooth curve of genus $g > 1$, then $|C(\mathbb{Q})|$ is finite.

This theorem is easy to state and beautiful, but giving a proof would require a full text book by itself. As our goal is to get the reader up to date on recent developments in isogeny-based cryptography, we must therefore sometimes skip depth and detail in proofs.

STAGES. The work on this introduction is divided into three stages.

- In the first stage, we describe the core material on elliptic curves and isogenies, and provide references for all other materials that is considered 'core material' for modern isogeny-based cryptography. This allows the reader to understand the basics of elliptic curves and isogenies that is necessary to read the references.
- In the second stage, we start to replace the references to 'core material' by our own writings on this material. This should simplify and unify the material for the reader. We furthermore add references to 'advanced material', which covers more niche subjects in isogeny-based cryptography.
- In the third stage, the work can be considered as a 'complete' introductory book on isogeny-based cryptography. However, as the field advances, we will need to keep sections up to date with developments, and 'advanced material' may become 'core material'.

Part 0

MATHEMATICAL FOUNDATIONS

MATHEMATICAL FOUNDATIONS

This part describes the mathematical foundations required to understand the other parts. This contains an overview of the necessary number theory, algebra, or other subjects.

§1 Basic Mathematics

TODO: add groups **TODO:** add finite abelian groups **TODO:** add CRT **TODO:** add basis of finite abelian groups

TODO: squares, non-squares in finite fields **TODO:** extensions of (finite) fields

TODO: add intersection of lines **TODO:** add resultant computation

Part 1

BASICS OF ISOGENY-BASED CRYPTOGRAPHY

BASICS OF ISOGENY-BASED CRYPTOGRAPHY

This part describes basic topics in isogeny-based cryptography, such as elliptic curves, isogenies, pairings, and the class group action.

1. Chapter I: Elliptic curves, curve models, torsion subgroups, arithmetic.
2. Chapter II: Elliptic Curves (Cryptography) Isogenies, Velu's formulas, isogeny graphs, endomorphisms, and representations.
3. Chapter III: Elliptic Curves (Cryptography) Pairings, the Weil pairing, the Tate pairing, Miller's algorithm, and cubical arithmetic.
4. Chapter IV: Elliptic Curves (Cryptography) Abelian varieties, Jacobians of hyperelliptic curves, Mumford representation, and genus 2.
5. Chapter V: Elliptic Curves (Cryptography) Higher-dimensional isogenies, Kani's lemma. **TODO:** test

I

ELLIPTIC CURVES

We start with mathematical aspects of elliptic curves, before we discuss cryptographical aspects and applications. We assume we work over a perfect field k, so that an algebraic closure $\overline{k}$ exists. When you are uncomfortable with fields as an abstract concept, just think of a finite field $\mathbb{F}_p$ or $\mathbb{F}_q$. For finite fields, we can think of $\overline{\mathbb{F}}_p$ as the union of all field extensions $\mathbb{F}_{p^k}$. Whenever we write q it is always a power of p, so that $\operatorname{char} \mathbb{F}_q$ is always p. Whenever we write ℓ, it is always assumed to be a prime different from p, except when explicitly stated $\ell = p$.

TODO: small todo list:

- fix Cref definition theorem mixup

§1 Elliptic Curves (Points)

The definition of an elliptic curve is the most central definition in isogeny-based cryptography.

Definition 1. An *elliptic curve* over a field k is a pair $(E, \mathbf{0})$, where E is a smooth projective curve of genus 1, and $\mathbf{0} \in E(k)$ a rational point. ◦

If this definition is not vague and abstract for you, you may likely skip all of Section 1. This section is mainly about the points on an elliptic curve.

Definition 2. The *rational points* on an elliptic curve is the set $E(k)$, and for any field extension K/k, we similarly denote by $E(K)$ the K-rational points of E. ◦

Unfortunately, both these definitions are almost entirely useless for concrete computations in this group. In practice, we are often given a curve in a given *model*, such as a *short Weierstrass curve*

$$E_{a,b} : y^2 = x^3 + ax + b \qquad a, b \in k, \tag{1}$$

a *Montgomery curve*

$$E_{A,B} : By^2 = x^3 + Ax^2 + x, \qquad A, B \in k, \tag{2}$$

or perhaps even a *twisted Edwards curve*

$$E_{a,d} : ax^2 + y^2 = 1 + dx^2y^2, \qquad a, d \in k. \tag{3}$$

So why does the abstract definition differ so much from the curve representations we use in practice? First, the abstract definition captures all of the curve models above, and resolves some difficulties that we encounter with the *affine* curve models and the mysterious 'point at infinity' $\mathbf{0} \in E(k)$. Second, each curve model above is mostly used in specific situations, because not every elliptic curve can be written as a curve in a given curve model, and because specific curve models have arithmetical properties that make them uniquely suitable to certain applications. This last point is crucial once we get to the cryptographic aspects of elliptic curves.

visualising elliptic curves. Having concrete equations helps us in visualising elliptic curves, which gives us the images we are so familiar with. For example, we may draw the curve $E : y^2 = x^3 - 2x + 4$ as follows.

Note: the picture we have drawn here displays the set of points (x, y) that satisfy $y^2 = x^3 - 2x + 4$ where $x, y \in \mathbb{R}$, so we have drawn $E(\mathbb{R})$! Through

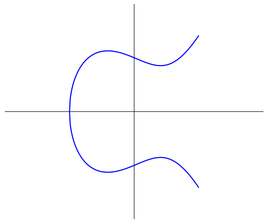

Figure 1: A visualisation over $\mathbb{R}$ of the elliptic curve $y^2 = x^3 - 2x + 4$.

magic, these visualisations often help our intuition for curves over finite fields too. But we should always be careful that the nature of elliptic curves over $\mathbb{R}$ is very different from the nature of curves over a finite field; we may borrow inspiration from such visualisations, but they may also misrepresent the actual situation we are working with. As an example, the picture of the same curve E over the finite fields $\mathbb{F}_7$ and $\mathbb{F}_{11}$ looks as follows.

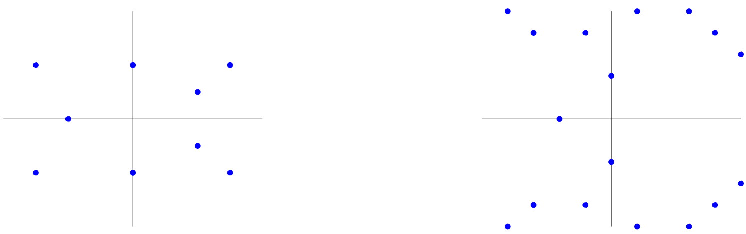

Figure 2: A visualisation over $\mathbb{F}_7$ and $\mathbb{F}_{11}$ of the elliptic curve $y^2 = x^3 - 2x + 4$.

Example 1. It is easy to compute the points on the above curves. One can check by a few finite field computations that $y^2 = x^3 - 2x + 4$ has

- 10 points over $\mathbb{F}_7$, namely $\mathbf{0}_E$ and the 9 affine points $(0,2)$, $(0,5)$, $(2,1)$, $(2,6)$, $(3,2)$, $(3,5)$, $(4,2)$, $(4,5)$, and $(5,0)$,
- 18 points over $\mathbb{F}_{11}$, namely $\mathbf{0}_E$ and the 17 affine points $(0,2)$, $(0,9)$, $(1,5)$, $(1,6)$, $(3,5)$, $(3,6)$, $(4,4)$, $(4,7)$, $(5,3)$, $(5,8)$, $(7,5)$, $(7,6)$, $(8,4)$, $(8,7)$, $(9,0)$, $(10,4)$, and $(10,7)$.

1.1 PROJECTIVE CURVES

By Definition 1, an elliptic curve is *projective*, but the curve models we have given so far are affine. We do so because affine curves are a bit easier to work with, practically, and are easier to visualise. Nevertheless, we should properly define elliptic curves as a projective variety. Luckily, this is not too difficult. We will show this for the affine short Weierstrass curve

$$E_{a,b} : y^2 = x^3 + ax + b.$$

This equation is in fact an affine chart of the 'proper' projective elliptic curve

$$\mathbf{E}_{a,b} : Y^2Z = X^3 + aXZ^2 + bZ^3,$$

whose points $P \in \mathbf{E}_{a,b}$ are tuples $(X : Y : Z) \in \mathbb{P}^3(k)$. These points are defined up to scalars, that is, $(X : Y : Z)$ is the same point as $(\lambda X : \lambda Y : \lambda Z)$ for any $\lambda \in k^*$.

Hence, we can simplify the points on $\mathbf{E}_{a,b}$ a bit: for any point $P \in \mathbf{E}_{a,b}(k)$ with $Z \neq 0$, we may normalize $(X : Y : Z)$ as $(\frac{X}{Z} : \frac{Y}{Z} : 1)$. But this then defines a point $(\frac{X}{Z}, \frac{Y}{Z})$ on the affine form of this curve $E_{a,b}(k)$, as it satisfies the curve equation. Vice versa, any point $(x, y) \in E_{a,b}(k)$ defines a point $(x : y : 1) \in \mathbf{E}_{a,b}(k)$.

The only difference between the points in $E_{a,b}(k)$ and $\mathbf{E}_{a,b}(k)$ must therefore be the points on $\mathbf{E}_{a,b}$ with $Z = 0$. But then, the left-hand side of the curve equation resolves to 0, and the right-hand side resolves to X^3, so we must have $X = 0$ too. This means such points are of the form $(0 : Y : 0)$ with $Y \in k^*$, and these are all equal to $(0 : 1 : 0)$ when scaled by Y^{-1}. Thus, there is only the single point $(0 : 1 : 0)$ that is well-defined for $\mathbf{E}_{a,b}$ but cannot be written as an affine point $(x, y) \in E_{a,b}$. This is the 'point at infinity' for $E_{a,b}$, which we denote $\mathbf{0}_E$.

IN PRACTICE. In practice, we may often simply use the affine curve $E_{a,b}$. Whenever a situation is unclear and involves $\mathbf{0}_E$, we will need to switch to the projective view $\mathbf{E}_{a,b}$, which gives us the complete correct view on $E_{a,b}$.

1.2 SMOOTH CURVES

By Definition 1, an elliptic curve is *smooth*, but what does that mean, practically speaking? We use our intuition over $\mathbb{R}$: the curve $C : y^2 = x^3$ is clearly *not* smooth.

The problem is the point $(0, 0)$, where C is quite literally very pointy. Although we clearly see this in the geometric picture, the mathematical community has collectively decided that such an approach to 'smoothness' is not rigorous enough. Instead, smoothness of a point on a curve can properly be established by non-vanishing differentials, which mathematically captures the 'pointiness'.

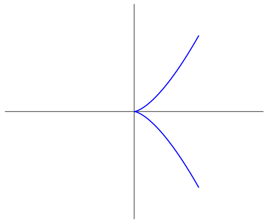

Figure 3: The curve $y^2 = x^3$ is not smooth at $(0,0)$.

Definition 3. Let C be a curve given by a non-constant polynomial $f(X_1, \ldots, X_n) = 0$, then $P \in C$ is a *singular point* when all derivates vanish, i.e.,

$$\frac{\partial f}{\partial X_1}(P) = 0, \quad \frac{\partial f}{\partial X_2}(P) = 0, \quad \ldots, \quad \frac{\partial f}{\partial X_n}(P) = 0.$$

If C has no singular points, we say that C is a *smooth curve*. ○

This gives us an easy way to verify that (the affine part of) a given curve is indeed smooth. We show this for a curve in short Weierstrass form.

Lemma 4. Let $E_{a,b}$ be a curve in short Weierstrass form $E_{a,b} : y^2 = x^3 + ax + b$. Then $E_{a,b}$ is a smooth curve if and only if $4a^3 + 27b^2 \neq 0$.

Proof. We first show all affine parts are smooth using the curve equation. The point $\mathbf{0}_E$ at infinity can be shown to be smooth using the projective curve equation. We get

$$\frac{\partial f}{\partial y} = 2y, \quad \frac{\partial f}{\partial x} = -3x^2 - a,$$

and so any potential singular point $P = (x_0, y_0)$ must have $y_0 = 0$. Thus, a singular point would have x_0 as a root of both the right-hand side $x^3 + ax + b$ and its derivative $-\frac{\partial f}{\partial x} = 3x^2 + a$. This only happens when the resultant (see Section 1) of these two polynomials is zero, and this resultant equals $4a^3 + 27b^2$. □

Similarly, a Montgomery curve $E_{A,B} : By^2 = x^3 + Ax^2 + x$ is a smooth curve when $A^2 \neq 4$ and $B \neq 0$. From now on, every time we describe a short Weierstrass curve or a Montgomery curve, we assume that they are elliptic curves and therefore smooth.

1.3 THE SIZE OF ELLIPTIC CURVES

The possible sizes and structures of elliptic curves over a field k are a fascinating subject. We first deal with possible sizes, before we discuss the possible structures. Although the size and structure of an elliptic curve over $E(\mathbb{Q})$, or more generally $E(K)$ for number fields $K/\mathbb{Q}$, is a fascinating subject, for cryptographic purposes we are interested in the size and structure over finite fields $\mathbb{F}_q$, with $q = p^k$. So, for now, we study $E(\mathbb{F}_q)$.

How large is $E(\mathbb{F}_q)$, roughly? Taking the short Weierstrass model as an example, we know there are only q^2 pairs $(x, y) \in \mathbb{F}_q \times \mathbb{F}_q$, and so at most we may get $q^2 + 1$ points (including the 'point at infinity'). But this is a rather crude and frankly ridiculous upper-bound. In reality, given any $x_0 \in \mathbb{F}_q$, the probability that $x_0^3 + ax_0 + b$ is a square should essentially be fifty-fifty, and so, roughly half of the q possible x_0 values should give us *two* corresponding points $(x_0, \pm\sqrt{x_0^3 + ax_0 + b}) \in E_{a,b}(\mathbb{F}_q)$. Adding the point at infinity $\mathbf{0}_E \in E(\mathbb{F}_q)$, which we always have, we should expect $|E(\mathbb{F}_q)|$ to be 'close' to $q + 1$.

Definition 5. For an elliptic curve E over $\mathbb{F}_q$, the *trace* t_q is the value

$$t_q := q + 1 - |E(\mathbb{F}_q)|. \tag{4}$$

$\circ$

We will see later (in **??**) why the name *trace* makes sense. For now, we may think of the trace as a measurement of how much larger or smaller $E(\mathbb{F}_q)$ is compared to the expected value of $q + 1$. It turns out that we have tight bounds on the possible sizes of $E(\mathbb{F}_q)$, or equivalently, the possible values of t_q, coming from 'deep mathematics' **TODO:** cite Hasse and Silverman.

Theorem 6. Let E be an elliptic curve over $\mathbb{F}_q$. Then $E(\mathbb{F}_q)$ satisfies

$$q + 1 - 2\sqrt{q} \;\leq\; |E(\mathbb{F}_q)| \;\leq\; q + 1 + 2\sqrt{q}. \tag{5}$$

We call this interval the *Hasse interval*. Equivalently, we get $|t_q| \leq 2\sqrt{q}$. $\circ$

If you are not amazed by this theorem the first time you see it, reflect on this result for a while. The result is extremely precise, giving only a narrow range of options for the size of $E(\mathbb{F}_q)$ relative to q, in a very simple formula. As we will see later, the result is optimal as the bounds are 'tight', as we can show that elliptic curves exist for every integer in the Hasse interval, up to very few very particular exceptions.

Example 2. We take as an example the prime $p = 163$ The Hasse interval gives us possible orders $139 \leq |E(\mathbb{F}_{163})| \leq 189$. Using MAGMA or sage, we can easily compute the orders of all possible short Weierstrass curves over $\mathbb{F}_{163}$. This gives us the following distribution.

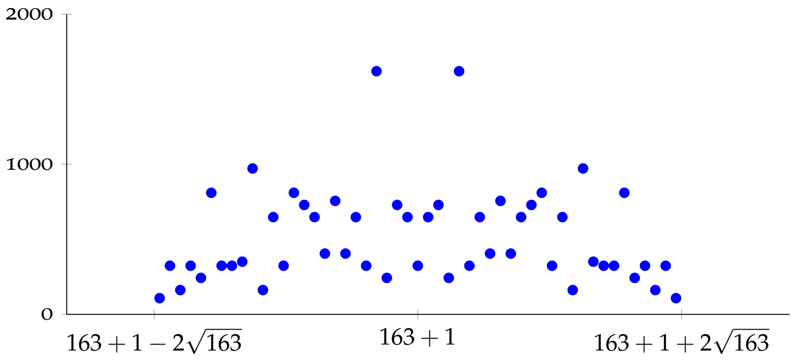

Figure 4: The number of Weierstrass curves of a given size over $\mathbb{F}_{163}$.

You will notice the symmetry around size $p + 1 = 164$ here, which has a simple explanation: we discuss this in the next section, Section 1.4.

1.4 THE TWIST OF AN ELLIPTIC CURVE

Everybody needs a friend. In stark contrast to the people who study them, every elliptic curve *has* a friend: its twist. We will define this here only for short Weierstrass curves and see a generalization later on.

Definition 7. Let $E_{a,b}$ be a short Weierstrass curve over $\mathbb{F}_q$ with $p > 2$, and let $d \in \mathbb{F}_q^*$ be a non-square. Then the elliptic curve $E_{a,b}^d$ is a *quadratic twist* of $E_{a,b}$, and is defined by

$$E_{a,b}^d : dy^2 = x^3 + ax + b.$$

A curve and its twist are not normal friends, but very *deep* friends, and this will show up in many places later. We get a first taste of this in the following result.

Lemma 8. Let $E_{a,b}$ be a short Weierstrass curve and $E_{a,b}^d$ a quadratic twist. Then

$$|E_{a,b}(\mathbb{F}_q)| + |E_{a,b}^d(\mathbb{F}_q)| = 2q + 2.$$

Equivalently, the trace of $E_{a,b}^d$ is the negation of the trace of $E_{a,b}$.

Proof. This is surprisingly simple: take any $x \in \mathbb{F}_q$, then we have the following options for $\beta = x^3 + ax + b = 0$.

- if $\beta = 0$, then $(x, 0)$ is a point for both $E_{a,b}$ and $E^d_{a,b}$,
- if β is non-zero and a square, then $(x, \sqrt{\beta})$ and $(x, -\sqrt{\beta})$ are two points on $E_{a,b}$,
- if β is non-zero and a non-square, then $(x, \sqrt{\beta/d})$ and $(x, -\sqrt{\beta/d})$ are two points on $E^d_{a,b}$.

Hence, in all cases we get 2 points for every $x \in \mathbb{F}_q$, either on $E_{a,b}$, or on $E^d_{a,b}$, or one on both. This gives us $2 \cdot q$ points, and adding $\mathbf{0}_E$ and $\mathbf{0}_{E^d}$ gives us a total of $2q + 2$ points. For the trace, let $t := q + 1 - |E_{a,b}(\mathbb{F}_q)|$ and $t^d := q = 1 - |E^d_{a,b}(\mathbb{F}_q)|$ then $t^d = -t$, because

$$t + t^d = 2q + 2 - |E_{a,b}(\mathbb{F}_q)| - |E^d_{a,b}(\mathbb{F}_q)| = 0.$$

□

Continuing our anthropomorphisation of elliptic curves, we find a symmetry in the Hasse interval: all curves have a friend who lives on the other side of the interval, except for a remarkable value in the middle of the interval, where $|E(\mathbb{F}_q)| = q + 1$.

Remark 1. We use the word twist to signify a quadratic twist, that is, an elliptic curve whose equation is "twisted" by a non-square. Generalizations sometimes exist, such as cubic twists or quartic twists, but we will not discuss these in detail.

1.5 SUPERSINGULAR ELLIPTIC CURVES

A few values for $|E(\mathbb{F}_q)|$, resp. t_q, are special. Clearly the value $|E(\mathbb{F}_q)| = q + 1$, resp. $t_q = 0$ is special, as it is precisely in the middle of the Hasse interval and matches with the 'rough expectation' sketched before.

Two other special values are the bounds $q + 1 \pm 2\sqrt{q}$. Curves that reach such a bound are called *minimal curves*, resp. *maximal curves*. When these bounds are integers, e.g. $q = p^{2k}$ for some k, such curves are very special, even. For $q = p^2$, this corresponds to curves of order $(p - 1)^2$ and $(p + 1)^2$, resp. trace $t_{p^2} = 2p$ and $t_{p^2} = -2p$. From the fact that curves with trace $-2p$, 0 and $2p$ all display special behaviour, we may guess that any multiple of p is special. Indeed, this is the case, but we will only find out how special later on. For now, we simply define these values as 'special'.

Definition 9. Let E be an elliptic curve over $\mathbb{F}_q$. If $t_q \equiv 0 \mod p$, then E is a *supersingular*[1] *elliptic curve*. Otherwise, E is an *ordinary elliptic curve*. ○

We should interpret this definition as follows: ordinary curves may come in very different shapes and sizes, as almost every integer in the Hasse interval corresponds to an ordinary curve. Supersingular curves, on the other hand, have very precisely described shapes and sizes, and are very peculiar in their behaviour.

1.6 EXISTENCE OF CURVES OF GIVEN SIZE.

For the existence of ordinary curves of a certain size, we could not hope for a better result than a result from Deuring and Waterhouse.

Theorem 10. For every integer n in the Hasse interval $[q+1-2\sqrt{q}, q+1+2\sqrt{q}]$ that is not a multiple of $p = \operatorname{char} \mathbb{F}_q$, there exists an (ordinary) elliptic curve E such that $|E(\mathbb{F}_q)| = n$.

Supersingular elliptic curves will be our core focus, as they are 'essentially' the only curves we use nowadays in isogeny-based cryptography. This is in large contrast with classical elliptic-curve cryptography, where we only use ordinary curves! We discuss this in a bit more detail in Section 4. The following result is very helpful when working with supersingular elliptic curves.

Lemma 11. Let E be a supersingular elliptic curve over $\mathbb{F}_q$. Then E is isomorphic to a curve defined over $\mathbb{F}_{p^2}$.

Proof. We show this later, in **??**, when we have developed more machinery. □

TODO: avoid annoying term isomorphic here

Again, this result should surprise you, as it is *a priori* very mysterious why we should be able to represent such curves over $\mathbb{F}_{p^2}$. Furthermore, this result holds more generally for a supersingular curve over *any* field of positive characteristic, not just finite fields. Because of Definition 11, let us more closely analyse the supersingular curves for the specific cases $q = p$ and $q = p^2$.

Lemma 12. Let E be a supersingular elliptic curve over $\mathbb{F}_p$ with $p \geq 5$. Then $|E(\mathbb{F}_p)| = p+1$ and $|E(\mathbb{F}_{p^2})| = (p+1)^2$. Furthermore, such a curve always exists.

1 The naming of these elliptic curves as *supersingular* is a historic blunder: it has confused students for decades as it has nothing to do with the usage of *singular* that we have used before. One should simply think of such curves as 'special' and if we could go back in time, we should have originally defined these curves as *special curves* instead of *supersingular*. In fact, when we later define what *superspecial* means, we will mean something even more special than *supersingular*. Alas, we are stuck with this name.

Proof. The result on the size is now easy: by Definition 6, we have $t_p \leq 2\sqrt{p}$. If $p \geq 5$, then $t_p = 0$ is the only multiple of p smaller than $2\sqrt{p}$. As $E(\mathbb{F}_p) \subseteq E(\mathbb{F}_{p^2})$, we must have $(p+1) \mid |E(\mathbb{F}_{p^2})|$. But E is still supersingular, so this is only possible for $|E(\mathbb{F}_{p^2})| = (p+1)^2$, resp. $t_{p^2} = -2p$. We leave the existence for a later moment. □

For $q = p^2$, the extremities of the Hasse interval are

$$q + 1 \pm 2\sqrt{q} = p^2 + 1 \pm 2p = (p \pm 1)^2,$$

so the minimal and maximal curves are both supersingular. By Definition 8, they are related by twists: the twist of a maximal curve is a minimal curve, and vice versa. Maximal curves over $\mathbb{F}_{p^2}$ are the supersingular curves that we use the most in isogeny-based cryptography. The other supersingular traces $t_{p^2} = 0, p, -p$ are more unusual, as the following result shows.

Lemma 13. Let $q = p^2$. A supersingular elliptic curve over $\mathbb{F}_{p^2}$ exists with

- trace $t_{p^2} = 0$, when $p \not\equiv 1 \mod 4$,
- trace $t_{p^2} = \pm p$, when $p \not\equiv 3 \mod 4$,
- trace $t_{p^2} = \pm 2p$, always.

SUMMARY

We may summarize this section as follows. The possible size of an elliptic curve E over a finite field $\mathbb{F}_q$ falls within the Hasse interval. A few values in the Hasse interval are special, namely the multiples of p. Elliptic curves of these sizes are *supersingular*, and are always isomorphic to a curve over $\mathbb{F}_{p^2}$. Elliptic curves of other sizes are *ordinary* curves.

- For all values n in the Hasse interval coprime to p, there exists on ordinary curve of size n.
- Supersingular curves over $\mathbb{F}_p$ with $p \geq 5$ are always of size $p+1$.
- Supersingular curves over $\mathbb{F}_{p^2}$ come in five different sizes, but we usually only work *maximal supersingular curves* of size $(p+1)^2$, and sometimes their twists, the *minimal supersingular curves*, of size $(p-1)^2$,

§2 Elliptic Curves (Arithmetic)

Elliptic curves are special because we can add points $P, Q \in E(k)$ together to get a well-defined point $R = P + Q \in E(k)$. This makes them the easiest example of an *abelian variety*, which we discuss in detail in Chapter IV. It would have made things much easier had we simply called them *abelian curves*. Alas, we seem to continuously misname things to confuse generations of students, but luckily, our superficial naming schemes will not affect the beauty of the underlying objects.

THE GROUP LAW, INFORMALLY. Most likely, you have seen something similar to the following to explain 'addition of points' on elliptic curves.

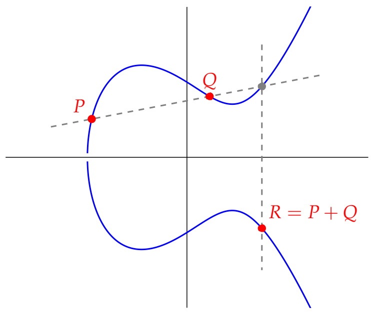

Figure 5: A visualisation over $\mathbb{R}$ of the addition of points on an elliptic curve.

This picture tries to say: if we take two random points $P, Q \in E(k)$, and we draw a line through them, they will intersect E in a third point. By reflecting this third point in the x-axis, we obtain a point $R \in E(k)$. If we define R as $P + Q$, this gives us a commutative group.

This should utterly amaze you. If it does not, reflect a bit longer on the construction. We draw seemingly random lines on our elliptic curve to derive $R = P + Q$, and somehow this construction *works*: If we have another point S, and we compute $R + S = P + Q + S$, we get the same point as if we first compute $T = P + S$ and then compute $T + Q = P + S + Q$. This is not at all clear from this geometric picture! Nor is it obvious why the result of such an addition would again have coordinates in the same field k as the original points. As an example, take another look at Figure 2, where we can use the same geometric construction using straight lines, but over finite fields. This is baffling! Unfortunately, a clear explanation relies on more advanced algebraic geometry that is out of scope, so for now, we simply have to accept the miracle of this group addition.

2.1 MORPHISMS, HOMOMORPHISMS AND ISOMORPHISMS

Even though we may have a different description for objects, sometimes they essentially 'feel' like the same object. We can make this rigorous by defining *isomorphisms* for such objects, so that two objects who are essentially the same in all the properties we care for but have a different description are *isomorphic*.

To do so, we first need to define what a morphism even is, and this is most properly done using the projective view on elliptic curves. So, we shortly think of an elliptic curve as a smooth curve of genus 1 defined by a homogeneous polynomial in X, Y, and Z.

Definition 14. Let E and E' be elliptic curves over k defined by homogeneous polynomials. A *morphism* of elliptic curves is a polynomial mapping between E and E' given by

$$\psi : (X : Y : Z) \mapsto (\psi_0(X,Y,Z), \psi_1(X,Y,Z), \psi_2(X,Y,Z))$$

with the ψ_i homogeneous polynomials of equal degree that satisfy the curve equation of E'.

Usually though, we simply look at morphisms on an affine patch, e.g., where $Z = 1$, and use the derived polynomials $\psi_i'(x,y) := \psi_i(X,Y,1)$. The above morphism then becomes a *rational map* between E and E' given by

$$\psi' : (x,y) \mapsto \left(\frac{\psi_0'(x,y)}{\psi_2'(x,y)}, \frac{\psi_1'(x,y)}{\psi_2'(x,y)} \right).$$

Definition 15. **TODO:** def homom as preserving group structure

In particular, any homomorphism φ satisfies $\varphi(\mathbf{0}_E) = \mathbf{0}_{E'}$. One can show that this is enough: any morphism that satisfies this is automatically a homomorphism. An isomorphism is then a special homomorphism, that not only preserves the elliptic curve, **TODO:** write nicer loses nothing of the essential structure of the elliptic curve, neither geometrically as a curve, nor algebraically as a group.

Definition 16. **TODO:** quick def of isomorphic

TODO: j-invariants for some curve models
TODO: as example isomorphic Montgomery curves
TODO: categorize all isomorphisms of short Weierstrass curves?
TODO: isomorphism of twists over larger field

2.2 EXPLICIT GROUP LAW COMPUTATIONS

We will make the group law explicit for short Weierstrass curves

$$E_{a,b} : y^2 = x^3 + ax + b$$

over k. The 'point at infinity' $\mathbf{0}_E$ will be the neutral element for the group operation, or in other words, $P + \mathbf{0}_E = P$ for any $P \in E(k)$. This explains why we chose $\mathbf{0}_E$ as the name for this point! Geometrically, we really can think of $\mathbf{0}_E$ as a point 'at infinity', which means that if we want to draw a line 'through' $\mathbf{0}_E$, we draw it vertically! A vertical line through $P \in E(k)$ will intersect E again in another point, and this point we can meaningfully define as $-P$. Figure 6 shows us this line through P and $-P$, hitting E at infinity[2] in the third point $\mathbf{0}_E$. Figure 7 shows us that we can also add points to themself, called *doubling*, by using the tangent line at a point Q. This makes sense compared to how we defined addition, as the tangent line intersects E twice in Q. We denote $Q + Q$ as $[2]Q$. The point P that intersects the x-axis is special: we get $[2]P = \mathbf{0}_E$.

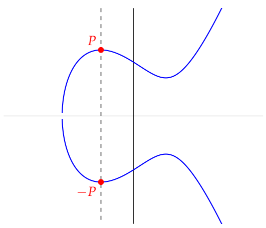

Figure 6: Negation of a point.

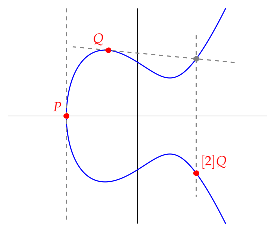

Figure 7: Doubling of points.

In terms of (x, y)-coordinates, we will write points as $P = (x_P, y_P)$. We find that negation only flips the y-coordinate, so $-P = (x_P, y_P)$. As it would be tedious to draw out elliptic curves geometrically every time we want to compute something on them, we now derive explicit formulas in terms of coordinates to compute addition of points.

ADDITION FORMULAS. From the geometric picture Figure 5, it is not difficult to derive the coordinates $R = (x_R, y_R)$ in terms of the coordinates $P = (x_P, y_P)$ and $Q = (x_Q, y_Q)$, assuming $x_P \neq x_Q$. We simply compute the line between P and Q, compute the third intersection $-R$ with E, and negate the y-coordinate to get R.

2 We have not drawn infinity because of a lack of space in our finite universe.

Let $L : y = \lambda x + \gamma$ denote the line through (x_P, y_P) and (x_Q, y_Q), then we can compute the *slope* λ as

$$\lambda = \frac{y_Q - y_P}{x_Q - x_P}. \tag{6}$$

By **??** and the fact that $x_R = x_{-R}$, we then get $\lambda^2 = x_P + x_Q + x_R$, so

$$x_R = \lambda^2 - x_P - x_Q. \tag{7}$$

To get the y-coordinate y_R, we can compute y_{-R} using the line L as $y_{-R} = \lambda x_{-R} + \gamma$. Although we did not compute γ, we have that $y_P = \lambda x_P + \gamma$ (and similarly for Q), and so $y_{-R} = \lambda x_{-R} + (y_P - \lambda x_P) = y_P - \lambda(x_P - x_{-R})$. Including the negation $y_R = -y_{-R}$ and $x_R = x_{-R}$, we finally get

$$y_R = \lambda(x_P - x_R) - y_P. \tag{8}$$

DOUBLING FORMULAS. To double $P = (x_P, y_P)$ we cannot use Equation (6) to compute the slop of the tangent line, but we can simply use the derivative $2y$ (with respect to y) and $3x^2 + a$ (with respect to x) to get the slope as

$$\lambda = \frac{3x_P^2 + a}{2y_P}. \tag{9}$$

The remaining derivation remains the same, giving us

$$x_{[2]P} = \lambda^2 - 2x_P, \qquad y_{[2]P} = \lambda(x_P - x_R) - y_P. \tag{10}$$

Example 3. **TODO:** take $p = 37$, and some curve, explicitly add and double some points

2.3 TORSION

We have already seen $[2]P$ as the doubling of a point P. Using repeated addition, we can easily generalize this to any multiple n of a point P as

$$[n]P := \underbrace{P + P + \cdots + P}_{n \text{ times}},$$

for any $n \in \mathbb{Z}$, using a negation if $n < 0$. This allows us to define the *order* of a point.

Definition 17. Let $P \in E(k)$. If $[n]P = \mathbf{0}_E$ for some integer n, then P is a *torsion point*. The smallest $n \in \mathbb{Z}_{>0}$ such that $[n]P = \mathbf{0}_E$ is the *order* of P. If no such n exists we say that P has infinite order.

Example 4. **TODO:** take Montgomery curve, compute torsion, explicitly discuss 2-torsion and factorization

Over an infinite field, such as $\mathbb{Q}$ or $\mathbb{R}$, it is not a priori clear if a point has finite order or not. Luckily, we focus on finite fields $\mathbb{F}_q$, and so, because $E(\mathbb{F}_q)$ is therefore finite, every point $P \in E(\mathbb{F}_q)$ must have some finite order. We may often want to look at the subgroup of $E(k)$ generated by a point.

Definition 18. Let $P \in E(k)$. Then $\langle P \rangle$ is the subgroup of $E(k)$ generated by P, i.e.

$$\langle P \rangle := \{P, [2]P, \ldots\}.$$

If P has order E, then $\langle P \rangle$ has size n.

TODO: flow here? add more?

A crucial subgroup of E is the n-torsion.

Definition 19. Let $n \in \mathbb{Z}_{>0}$. The *n-torsion* of an elliptic curve E over k is the subgroup of $E(\bar{k})$ defined by

$$E[n] := \{P \in E(\bar{k}) \mid [n]P = \mathbf{0}_E\}.$$

We may think of $[n]$ as a map

$$[n] : E(\bar{k}) \to E(\bar{k}), \quad P \mapsto P + P + \cdots + P,$$

so that $E[n] = \ker[n]$. This map is called the *multiplication-by-n* map. Two remarks are in place.

1. The n-torsion is *not* the collection of points of order n. For example, if $n = 6$, then a point P of order 3 satisfies $[3]P = \mathbf{0}$ and therefore also $[6]P = [2]\mathbf{0} = \mathbf{0}$, so $P \in E[6]$. Thus, you should think of $E[n]$ as the subgroup of all points of order d where $d \mid n$. This always includes $\mathbf{0}$, as $\mathbf{0}$ has order $d = 1$, and 1 always divides n.

2. This is the first time we really needed the algebraic closure $\bar{k}$, and you may wonder why we did not define $E[n]$ as the subgroup of points $P \in E(k)$ with $[n]P = \mathbf{0}$ instead. We denote this last group as $E[n](k)$, so

$$E[n](k) := \{P \in E(k) \mid [n]P = \mathbf{0}_E\}.$$

When these groups coincide, in other words, when $E[n] \subseteq E(k)$, we say that the n-torsion is k-rational, or simply *rational* when it is obvious which field is implied. One should think of $E[n]$ as an abbreviation of $E[n](\overline{k})$.

There are a number of very good reasons to define $E[n]$ over $\overline{k}$, but the following lemma may be the clearest one for now.

Lemma 20. Let E be an elliptic curve over k and take $n \in \mathbb{Z}_{>0}$ coprime to $\operatorname{char} k$. Then

$$E[n] \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

Proof. We need more machinery for this, so we postpone this to Chapter II. □

This is a very elegant and useful result. For finite fields, we may think of this in the following way: Let E be an elliptic curve over $\mathbb{F}_q$, and n any positive integer coprime to p. Even though our curve may or may not have points of order n, we know that if we extend to some (potentially very very large, but still finite) field $\mathbb{F}_{p^k}$, we eventually obtain $E[n] \subseteq E(\mathbb{F}_{p^k})$.

Example 5. **TODO:** continue previous example, by showing that the 2-torsion lives in the first extension. Then look for the full 3-torsion or something.

We now take a closer look at the rational n-torsion over finite fields, i.e., $E[n](\mathbb{F}_q)$ for a curve E over $\mathbb{F}_q$ for the specific case where $n = \ell$ is a prime different from $p = \operatorname{char} \mathbb{F}_q$.

Lemma 21. Let E be an elliptic curve over $\mathbb{F}_q$, and take a prime $\ell \neq p$. Then the rational ℓ-torsion $E[\ell](\mathbb{F}_q)$ is either

1. rank 0, which means $E(\mathbb{F}_q)$ has no points of order ℓ, so $E[\ell](\mathbb{F}_q) = \{\mathbf{0}_E\}$.
2. rank 1, which means $E[\ell](\mathbb{F}_q) \cong \mathbb{Z}/\ell\mathbb{Z}$. Thus, it is generated by a single non-zero point $P \in E[\ell](\mathbb{F}_q)$, i.e.,
$$E[\ell](\mathbb{F}_q) = \langle P \rangle = \{P, [2]P, \ldots, [\ell - 1]P, \mathbf{0}_E\}.$$
3. rank 2, which means $E[\ell](\mathbb{F}_q) = E[\ell] \cong \mathbb{Z}/\ell\mathbb{Z} \times \mathbb{Z}/\ell\mathbb{Z}$. Thus, it is generated by two non-zero points $P, Q \in E[\ell]$ such that $Q \neq [\lambda]P$ for any $\lambda \in \mathbb{Z}_{>0}$. That is,
$$E[\ell](\mathbb{F}_q) = \langle P, Q \rangle = \{[a]P + [b]Q, \quad a, b \in [1 \ldots \ell]\}.$$

Proof. From Definition 20, we see that the ℓ-torsion clearly cannot be larger than rank 2. Now, if there are no points of order ℓ, we obtain the rank 0 case. If there is at least a single point P of order ℓ, this immediately generates the subgroup $\langle P\rangle \subseteq E[\ell](\mathbb{F}_q)$ of rank 1. If $E[\ell](\mathbb{F}_q)$ is still larger than $\langle P\rangle$, then there is some non-zero $Q \in E[\ell](\mathbb{F}_q)$ with $Q \notin \langle P\rangle$, i.e., linearly independent. Then P and Q together generate $E[\ell]$, and as both are elements of $E(\mathbb{F}_q)$, we get the rank-2 case $E[\ell](\mathbb{F}_q) = E[\ell]$. □

Similar to how the field size $|\mathbb{F}_q|$ imposes strong conditions on $|E(\mathbb{F}_q)|$, the structure of $\mathbb{F}_q^*$ imposes conditions on the structure of $E(\mathbb{F}_q)$. This is, once again, a connection arising from 'deep mathematics'. Nevertheless, we have a clear example when the n-torsion is rational.

Lemma 22. Let E be an elliptic curve over $\mathbb{F}_q$ with $E[n] \subseteq E(\mathbb{F}_q)$ for some $n \in \mathbb{Z}_{>0}$ coprime to p. Then $n \mid q-1$.

Proof. Two proofs for this both rely on material that we will discuss a bit later. First, we will use the characteristic polynomial of Frobenius in Chapter II, and second, we will use the non-degeneracy of the Weil pairing, which we introduce in Chapter III, to show $\mu_n \subseteq \mathbb{F}_q^*$, which implies $n \mid q-1$. □

p-TORSION. You may wonder why we have assumed n coprime to p so far to talk about n-torsion. As always, when working over a field of positive characteristic p, objects involving p are always a bit stranger. This is also the case for $E[p]$ for elliptic curves over k with $\operatorname{char} k = p$.

Lemma 23. Let E be an elliptic curve over k with $\operatorname{char} k = p$. Then, for any integer $r > 0$,

- if E is ordinary, then $E[p^r] \cong \mathbb{Z}/p^r\mathbb{Z}$,
- if E is supersingular, then $E[p^r] \cong 0$.

Remarkably, we can distinguish ordinary curves and supersingular curves by the existence of points of order p! Furthermore, Definition 23 is more general than our original definition of supersingularity in Definition 9, which only holds for curves over finite fields. One may in fact use Definition 23 as the defining property of supersingularity, and deduce Definition 9 as a consequence.

SYLOW TORSION. A slight generalization of the n-torsion is the Sylow-n torsion, which essentially consists of all points $P \in E(\mathbb{F}_q)$ such that the order of P is *some power* of n.

Definition 24. Let E be an elliptic curve over $\mathbb{F}_q$, and let $n \in \mathbb{Z}_{>0}$. The *Sylow-n torsion* of E over $\mathbb{F}_q$, denoted $\mathcal{S}_n(E)$, is the subgroup

$$\mathcal{S}_n(E) := \{P \in E(\mathbb{F}_q) \mid [n^k]P = \mathbf{0}_E \text{ for some } k \in \mathbb{Z}_{>0}\}.$$

This subgroup appears under a few different names and notations, such as $E[n^\infty](\mathbb{F}_q)$ or 'the rational $n^\bullet$-torsion'. Because of the Chinese Remainder Theorem (**??**), we can always decompose the Sylow-n torsion into its prime divisors. Therefore, we are mostly interested in the Sylow-ℓ torsion, for ℓ prime.

Lemma 25. Let E be an elliptic curve over $\mathbb{F}_q$, and let ℓ be a prime different from p. Then, the rank of $\mathcal{S}_\ell(E)$ is equal to the rank of $E[\ell](\mathbb{F}_q)$, and

$$\mathcal{S}_\ell(E) \cong \mathbb{Z}/\ell^f\mathbb{Z} \times \mathbb{Z}/\ell^g\mathbb{Z},$$

with $f, g \in \mathbb{N}$ and $f \geq g$. When $f = g > 0$, we say that $\mathcal{S}_\ell(E)$ is *symmetric*.

Proof. For a basis (P, Q) of $E[\ell](\mathbb{F}_q)$, we get $P, Q \in \mathcal{S}_\ell(E)$ by definition, so the rank of $\mathcal{S}_\ell(E)$ is at least the rank of $E[\ell](\mathbb{F}_q)$. Vice versa, any point $P \in \mathcal{S}_\ell(E)$ has a multiple in $E[\ell](\mathbb{F}_q)$ and so the rank is at most the rank of $E[\ell](\mathbb{F}_q)$, hence equal. As $\mathcal{S}_\ell(E)$ is an abelian group whose size is some power of ℓ, with rank bounded by 2, it must be isomorphic to $\mathbb{Z}/\ell^f\mathbb{Z} \times \mathbb{Z}/\ell^g\mathbb{Z}$ for some $f, g \in \mathbb{N}$. $\square$

Example 6. **TODO:** take somewhat random prime until it has like $3^3 \times 3$ torsion?

2.4 GROUP STRUCTURE

We can now finally tackle the possible group structures of the finite abelian group $E(\mathbb{F}_q)$. This is remarkably elegant, and a key theorem that should always be in the back of your mind.

Theorem 26. Let E be an elliptic curve over $\mathbb{F}_q$. Then

$$E(\mathbb{F}_q) \cong \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z},$$

for some $M, N \in \mathbb{N}$ with $N \mid M$.

Proof. Let n be the size of $E(\mathbb{F}_q)$. Then for all prime divisors $\ell \mid n$, by Definition 25 the Sylow-ℓ torsion is of the form $\mathbb{Z}/\ell^{f_\ell}\mathbb{Z} \times \mathbb{Z}/\ell^{g_\ell}\mathbb{Z}$. By the Chinese Remainder Theorem, we get

$$\begin{aligned} E(\mathbb{F}_q) &\cong \prod_{\ell \mid n} \mathcal{S}_\ell(E) \\ &\cong \prod_{\ell \mid n} \mathbb{Z}/\ell^{f_\ell}\mathbb{Z} \times \mathbb{Z}/\ell^{g_\ell}\mathbb{Z} \\ &\cong \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z} \end{aligned}$$

for $M = \prod \ell^{f_\ell}$ and $N = \prod \ell^{g_\ell}$. Note that we include $\ell = p$ here to deal with $E[p^r] \cong \mathbb{Z}/p^r\mathbb{Z}$ for ordinary curves. $\square$

Remark 2. From $E(\mathbb{F}_q) \cong \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$ with $N \mid M$, it follows that the whole N-torsion $E[N]$ is rational. Hence, by Definition 22, we get that $N \mid q-1$. This implies strong conditions on when curves can have such a shape. For example, *any* elliptic curve over $\mathbb{F}_p$ with $p = 23$ can only potentially have rational 2-torsion and/or rational 11-torsion.

Example 7. Take random curves over random primes to derive some group structures

We now wonder about the reverse question: given $\mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$ with $N \mid M$, when does an elliptic curve over $\mathbb{F}_q$ actually exist with this group structure? As you may expect, the answer is the best answer we could hope for, although we have to talk about ordinary and supersingular curves separately.

GROUP STRUCTURE OF ORDINARY CURVES. For an ordinary curve, we know that $E(\mathbb{F}_q)$ must be of the form $\mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$, with $|E(\mathbb{F}_q)| = M \cdot N$ in the Hasse interval and coprime to p. Furthermore, we observed that we must have $N \mid q-1$. By Deuring-Waterhouse (Definition 10), we know that there always exists a curve of the right *size* $M \cdot N$. A follow-up result by Tsfasman, Voloch, and Rück shows that there always exists a curve of the right *structure* too.

Theorem 27. Let $M, N \in \mathbb{N}$ with $N \mid M$ and $N \mid q-1$. If $M \cdot N$ is in the Hasse interval and coprime to p, then there exists an ordinary elliptic curve E over $\mathbb{F}_q$ such that

$$E(\mathbb{F}_q) \cong \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}.$$

For ordinary curves, this is essentially the best result we could hope for: we are given strong conditions on M and N by Definition 6 and Definition 22, and we get the existence of a curve E for every M and N that satisfy those conditions. In other words, we get the following mantra.

The Everything Mantra.

For ordinary curves, everything that can exist, must exist.

GROUP STRUCTURE OF SUPERSINGULAR CURVES. Supersingular curves have to be a bit more special, of course. We separate the two cases $q = p$ and $q = p^2$.

Lemma 28. Let E be a supersingular curve over $\mathbb{F}_p$ with $p > 5$. Then

$$E(\mathbb{F}_p) \cong \mathbb{Z}/(p+1)\mathbb{Z} \quad \text{or} \quad E(\mathbb{F}_p) \cong \mathbb{Z}/\left(\frac{p+1}{2}\right)\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}.$$

Proof. We know $|E(\mathbb{F}_p)| = p + 1$ by Definition 12. If some n-torsion $E[n]$ is rational, then $n \mid p - 1$ by Definition 22. But then n divides $p - 1$ and $p + 1$, so $n = 2$. By Definition 26, we can then only have $N = 1$ or $N = 2$, from which the result follows. □

For $q = p^2$, our main case of interest are the minimal supersingular curves of order $(p-1)^2$ (resp. $t_{p^2} = 2p$) and maximal supersingular curves of order $(p+1)^2$ (resp. $t_{p^2} = -2p$).

Lemma 29. Let E be a supersingular curve over $\mathbb{F}_{p^2}$. If E is minimal, that is, $|E(\mathbb{F}_{p^2})| = (p-1)^2$, then

$$E(\mathbb{F}_{p^2}) \cong \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z}.$$

If E is maximal, that is, $|E(\mathbb{F}_{p^2})| = (p+1)^2$, then

$$E(\mathbb{F}_{p^2}) \cong \mathbb{Z}/(p+1)\mathbb{Z} \times \mathbb{Z}/(p+1)\mathbb{Z}.$$

Proof. This is postponed to Chapter II, when we have met Frobenius and its characteristic polynomial. □

The other cases exhibit similar behaviour to ordinary curves, in the sense that everything that can occur, up to the strong conditions we have sketched before, will occur. The case of minimal and maximal supersingular curves is therefore very peculiar, with this specific group structure strongly enforced by deeper reasons. This will turn out to be a key connection to why we use these curves specifically in isogeny-based cryptography.

2.5 CURVE MODELS

So far, we have either discusses elliptic curves abstractly, as defined in Definition 1, or sometimes using examples in the short Weierstrass model $y^2 = x^3 + ax + b$. We have only very briefly touched on some other curve models. In this section, we will discuss different curve models, when a curve can be written in a certain model, and why you would want to use one model or another.

In general, any elliptic curve E over any field k can be written in *long Weierstrass form*

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in k.$$

Proving this requires a bit of non-trivial algebraic geometry, up to the Riemann-Roch theorem, but we will not go into these details here. The long Weierstrass form is helpful, but fairly annoying and expensive for arithmetic: there are simply too many coefficients[3] a_i to work with.

Luckily, in practice, we can almost always work with more convenient curve models. We present the two that are most commonly used in isogeny-based cryptography.

SHORT WEIERSTRASS CURVES. We introduced the short Weierstrass curve in Equation (1). It is aptly named, as it is indeed shorter than the long Weierstrass form. It turns out that almost every elliptic curve can also be written in this shorter Weierstrass form, as long as we are not working over a field of characteristic 2 or 3. As isogeny-based cryptography almost always works with curves of large characteristic, we can essentially always assume that our curve has a short Weierstrass model.

Lemma 30. Let E be an elliptic curve over k, with $\operatorname{char} k \neq 2, 3$. Then, there exist $a, b \in k$ such that E is isomorphic to a *short Weierstrass curve*

$$E_{a,b} : y^2 = x^3 + ax + b.$$

MONTGOMERY CURVES. The curve model that you will see in almost every paper in isogeny-based cryptography is the Montgomery curve that we introduced in Equation (2) as

$$E_{A,B} : By^2 = x^3 + Ax^2 + x,$$

with $B \neq 0$ and $A^2 \neq 4$. It became well-known for its fast arithmetic using the so-called *Montgomery ladder*, which we discuss in Section 3. Fast elliptic curve cryptography relies on fast ladders, and similarly, isogeny-based cryptography inherits this fast arithmetic by using Montgomery curves.

Already from the defining equation, we notice that not every elliptic curve E can be isomorphic to a curve in Montgomery form: We have the point $(0, 0) \in$

3 One may wonder why the naming convention of these coefficients is so odd, and why we skip a_5. For algebro-geometric reasons, we want to assign weight 2 to the coordinate x and weight 3 to the coordinate y. By giving the i-th coefficient a_i weight i, this naming convention ensures each term has weight 6.

$E_{A,B}(\mathbb{F}_q)$ for any $A, B \in k$, and this is a point of order 2. So certainly curves without rational 2-torsion, that is, $E[2](\mathbb{F}_q) = 0$, will not have a Montgomery model. With a bit more algebraic manipulation, we can even show that $4 \mid |E_{A,B}(\mathbb{F}_q)|$ [Mon87].

The key criterion is closely related, and at its core comes down to the behavior of addition by $(0,0)$ on a Montgomery curve.

Lemma 31. Let $E_{A,B}$ be a Montgomery curve over $\mathbb{F}_q$. Let $P = (x_P, y_P) \in E(\mathbb{F}_q)$ with $P \neq (0,0)$. Then the x-coordinate of $P + (0,0)$ is $\frac{1}{x_P}$.

Proof. We have not derived addition formulas on Montgomery curves explicitly, but they are easy to derive (as an exercise). From this, we quickly derive the result by adding P and $(0,0)$ using the addition formulas. □

This is the crucial behavior that we need from E to have an isomorphic curve $E_{A,B}$ in Montgomery form: A point $TinE[2](\mathbb{F}_q)$, that will map to $(0,0) \in E_{A,B}$ so that the resulting addition $P \mapsto P + (0,0)$ maps $x_P \mapsto 1/x_P$. We can describe this condition algebraically [OKS00].

Lemma 32. Let $E_{a,b}$ be an elliptic curve in short Weierstrass form. Then there exists an isomorphic Montgomery curve $E_{A,B}$ if and only if there are $\alpha, \beta \in \mathbb{F}_q$ such that

$$\alpha^3 + a \cdot \alpha + b = 0, \quad \text{and} \quad 3\alpha^2 + a = \beta^2.$$

Proof. We follow Costello and Smith [CS18]. The first condition ensures we have a point $(\alpha, 0) \in E_{a,b}$ that we can send to $(0,0)$. When we do this via the isomorphism $(u,v) \mapsto (x,y) = ((u-\alpha)/\beta, v/\beta)$, the second condition ensures that addition by $(0,0)$ acts as $x_P \mapsto 1/x_P$. □

The reverse is always true by Definition 30: any Montgomery curve over a field of characteristic $p \neq 2, 3$ is isomorphic to a short Weierstrass curve.

SIMPLE MONTGOMERY CURVES. In many situations, we only deal with Montgomery curves of the form

$$E_A : y^2 = x^3 + Ax^2 + x, \qquad A \in k,$$

that is, setting $B = 1$. We then refer to A as the *Montgomery coefficient*. It turns out this gives us exactly the maximal supersingular curves over $\mathbb{F}_{p^2}$, assuming $p > 3$ and $p \equiv 3 \mod 4$.

Lemma 33. Let $p > 3$ and $p \equiv 3 \mod 4$, and let E be a supersingular curve over $\mathbb{F}_{p^2}$ of order $(p+1)^2$, then E is isomorphic to E_A for some $A \in \mathbb{F}_{p^2}$.

Vice verse, if E_A is a supersingular Montgomery curve with $A \in \mathbb{F}_{p^2}$, then $|E_A(\mathbb{F}_{p^2})| = (p+1)^2$.

Proof. **TODO:** improve We need a bit of magic here: for any supersingular Montgomery curve, $A+2$ and $A-2$ are squares in $\mathbb{F}$. Now, if we double the point P with $x_P = 1$, we can compute that $x_{[2]P} = 0$, and so $[2]P$ must be $(0,0)$, so P is a point of order 4. Only maximal supersingular curves, with group structure $\mathbb{Z}/(p+1)\mathbb{Z} \times \mathbb{Z}/(p+1)\mathbb{Z}$ have points of order 4, so P with $x_P = 1$ is on E if and only if E is a maximal supersingular curve. The associated y-coordinate on a Montgomery curve satisfies $By^2 = A+2$, and as $A+2$ is square, B must be square too. But then we can take the isomorphism $(x,y) \mapsto (x, \frac{1}{\sqrt{B}}y)$ to get an isomorphic Montgomery curve with $B = 1$.

Vice versa, a supersingular Montgomery curve E_A has a point $P = (1, \sqrt{A+2}) \in E[4](\mathbb{F}_{p^2})$, and so the group structure must be $\mathbb{Z}/(p+1)\mathbb{Z} \times \mathbb{Z}/(p+1)\mathbb{Z}$. $\square$

Remark 3. Rather surprisingly, I cannot find an elementary proof that $A+2$ and $A-2$ are squares for supersingular Montgomery curves, and so, a proper proof is delayed until at least Chapter II.

Over $\mathbb{F}_p$, the situation is slightly more complicated. Although any supersingular Montgomery curve E_A with $A \in \mathbb{F}_p$ must have order $p+1$ by Definition 12, the converse is not true: not every supersingular curve over $\mathbb{F}_p$ of order $p+1$ can be written with $B = 1$. For more details, see [**DBLP:conf/pqcrypto/CastryckD20**, CS18].

SUMMARY

Arithmetic on elliptic curves (in Weierstrass form) follows easily from the geometric view. Over finite fields, any elliptic curve E has n-torsion $E[n] \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$, although this torsion most likely lives in an extension field. The curve itself has a group structure over $\mathbb{F}_q$ of the form

$$E(\mathbb{F}_q) \cong \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z},$$

with $N \mid M$ and $N \mid q-1$. For *ordinary* curves, we have the *Everything Mantra*: any group structure that can potentially happen within the constraints, must happen. For *supersingular* curves over $\mathbb{F}_p$, we have two possible group structures, related to the 2-torsion. For *supersingular* curves over $\mathbb{F}_{p^2}$, we care about minimal and maximal curves, whose group structures must be $\mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z}$, resp. $\mathbb{Z}/(p+1)\mathbb{Z} \times \mathbb{Z}/(p+1)\mathbb{Z}$. When $p \equiv 3 \mod 4$ and $p > 3$, which is almost always the case for later applications, these maximal supersingular curves can be represented in (short) Montgomery form

$$E_A : y^2 = x^3 + Ax^2 + x, \qquad A \in \mathbb{F}_{p^2}.$$

§3 The Kummer Line

In the very first article introducing elliptic curves in cryptography, **miller-ecc** notes that one could perform most operations required for elliptic curve cryptography using only the x-coordinate of points $P \in E(k)$. In particular for curves in Montgomery form, such x-only approaches are significantly faster. This holds not only for classical ECC, but also for current-day isogeny-based cryptography.

As the x-coordinate of a point is equal for both P and $-P$, mathematically, working with x-only arithmetic means an identification of these two. That is, we work on E with equivalence between P and $-P$. This turns out to be an algebraic variety $K_E = E/\langle \pm 1 \rangle$ isomorphic to the projective line $\mathbb{P}^1$, named the Kummer line of E. The Kummer line K_E does not inherit the exact group structure of E due to the identification of P and $-P$. Nevertheless, K_E is a pseudo-group, which means we can still perform scalar multiplication and differential addition. This turns out to be enough for most cryptographic applications, such as a Diffie-Hellman key exchange based on the group $E(k)$, as it only requires the computation of $P \mapsto [n]P$ for $n \in \mathbb{Z}$.

RESOURCES

Costello and Smith [CS18] have written almost all you need to know on x-only arithmetic on Montgomery curves. More advanced resources are **[HR19, cryptoeprint:2025/543]**

TODO: expand

§4 Elliptic Curves (Cryptography)

The usage of elliptic curves in cryptography long predates isogeny-based cryptography. **koblitz-ecc** and **miller-ecc** independently suggested using elliptic curves over finite fields as a potential replacement for finite field Diffie-Hellman. Out of the thousands of papers written on this subject, we will try to present a quick summary in a few paragraphs.

The main idea is simple: the group $E(k)$ is abelian, and so $[ab]P = [ba]P$ for integers $a, b \in ZZ$. So if we take for P some point on $E(k)$ and share it publicly, Alice takes a secret value $a \in \mathbb{Z}$, Bob takes a secret value $b \in \mathbb{Z}$, and they exchange $A := [a]P$ and $B := [b]P$ with each other (their public keys), then they can both derive $K = [ab]P$, Alice by computing $[a]B = [ab]P$, and Bob by $[b]A = [ba]P$. The assumption here is that no one else can compute K as long as they do not have the secrets a and b. More precisely, we are assuming the hardness of two problems.

Problem 1 (Discrete Logarithm)**.** Let $P \in E(k)$ and $a \in \mathbb{Z}$. Given P and $A = [a]P$, compute a.

Problem 2 (Computational Diffie-Hellman)**.** Let $P \in E(k)$ and $a, b \in \mathbb{Z}$. Given P, $A = [a]P$, and $B = [b]P$, compute $K = [ab]P$.

It is easy to see that if you can solve the Discrete Logarithm problem, then you can easily solve the Computational Diffie-Hellman problem.

TODO: hardness of CDH

TODO: hardness of discrete logarithm

TODO: example of curve25519

II

ISOGENIES

isogenies placeholder

III

PAIRINGS

pairings placeholder

IV

ABELIAN VARIETIES

abelian varieties placeholder

TODO: Milne's mantra **TODO:** make very special/not very special joke

V

HIGHER-DIMENSIONAL ISOGENIES

We assume basic familiarity with the following material:

- Chapter I:
- Chapter II: Elliptic Curves (Cryptography)

higher-dimensional isogenies

Part 2

ADVANCED TOPICS IN ISOGENY-BASED CRYPTOGRAPHY

ADVANCED TOPICS IN ISOGENY-BASED CRYPTOGRAPY

This part describes advanced topics in isogeny-based cryptography, such as the Deuring correspondence, higher-dimensional isogenies, and generalized class group actions.

Part 3

CRYPTOGRAPHIC CONSTRUCTIONS

CRYPTOGRAPHIC CONSTRUCTIONS

This part describes the cryptographic constructions, such as SQIsign or CSIDH, that are built on top of Parts 1 and 2.

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